Magnet Axiom Android Analysis

Daniel Forster

Pace University; IT-664

Investigation Number – Case 1

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Executive Summary

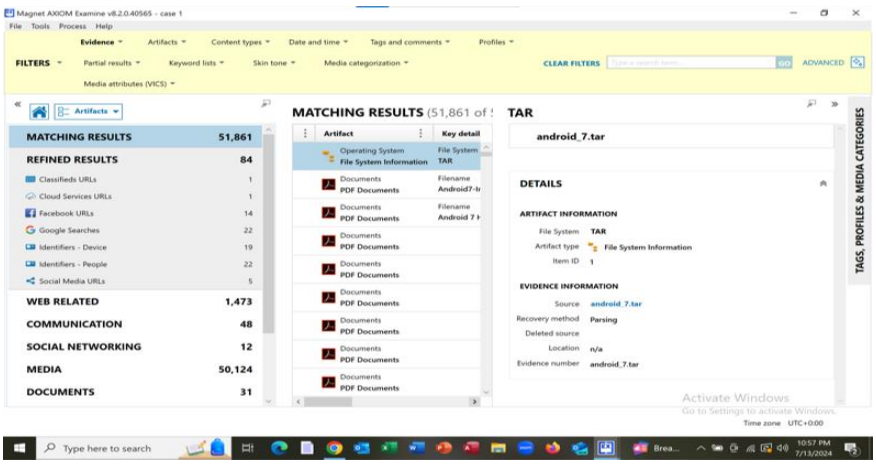
This device is being examined based on a project for the IT-664 class at Pace University. This project is being performed using the Magnet Axiom suite, which involves Magnet Axiom Process and Magnet Axiom Examine, two applications widely used in computer forensics. Law enforcement agencies and private entities mainly use Magnet Axiom. In this investigation, we used Magnet Axiom to extract data from this individual's entire Android phone, where social media, messaging, photo, and Personally Identifiable Information (PII) artifacts were obtained. I, Daniel Forster, am the sole investigator of this investigation.

Biography

My name is Daniel Forster, and I am the investigator in this case. I have been studying in this field for five years. I graduated from Assumption University in May 2023 with a Bachelor of Science in cybersecurity. During this time, I have had experience with tools and operating systems used in this field, such as Autopsy, Wireshark, Kali Linux, Windows, and Mac OS. I am currently pursuing my master's degree at Pace University, and I am currently about to complete my first year in the program. During this time, I gained more knowledge in cybersecurity and computer forensics. The tools that I used this past school year were Wireshark, Magnet Axiom, and different programs on Kali Linux.

The Methodology

The methods of this investigation involved an Intel-based 2019 MacBook Pro and the Magnet Axiom Suite, which involved the Magnet Axiom Process and the Magnet Axiom Examine. The MacBook Pro utilized the prebuilt software assistant Bootcamp to install Windows onto the MacBook. Bootcamp allows a Mac to run both Windows and Mac OS on the operating system by splitting the Mac's hard storage disk into two parts, allowing a Windows section (known as a Windows partition) from the main Mac disk storage drive. This action was necessary because Magnet Axiom is a Windows-based application, and the current device available was this Intel 2019 MacBook Pro. Magnet Axiom was used for this investigation due to its ability to analyze digital devices. This software does a superior job of analyzing digital hardware in computers, cellular devices, car systems, and IoT devices. For this investigation, this device was analyzed by Magnet Axiom, which analyzed a Physical Data Acquisition of the device. Physical data acquisition is where an investigator creates a bit-by-bit copy (or an identical image) of the smartphone, which is then processed by Magnet Axiom, which breaks the contents of the phone into artifacts where an Investigator can quickly analyze a device, seen below.



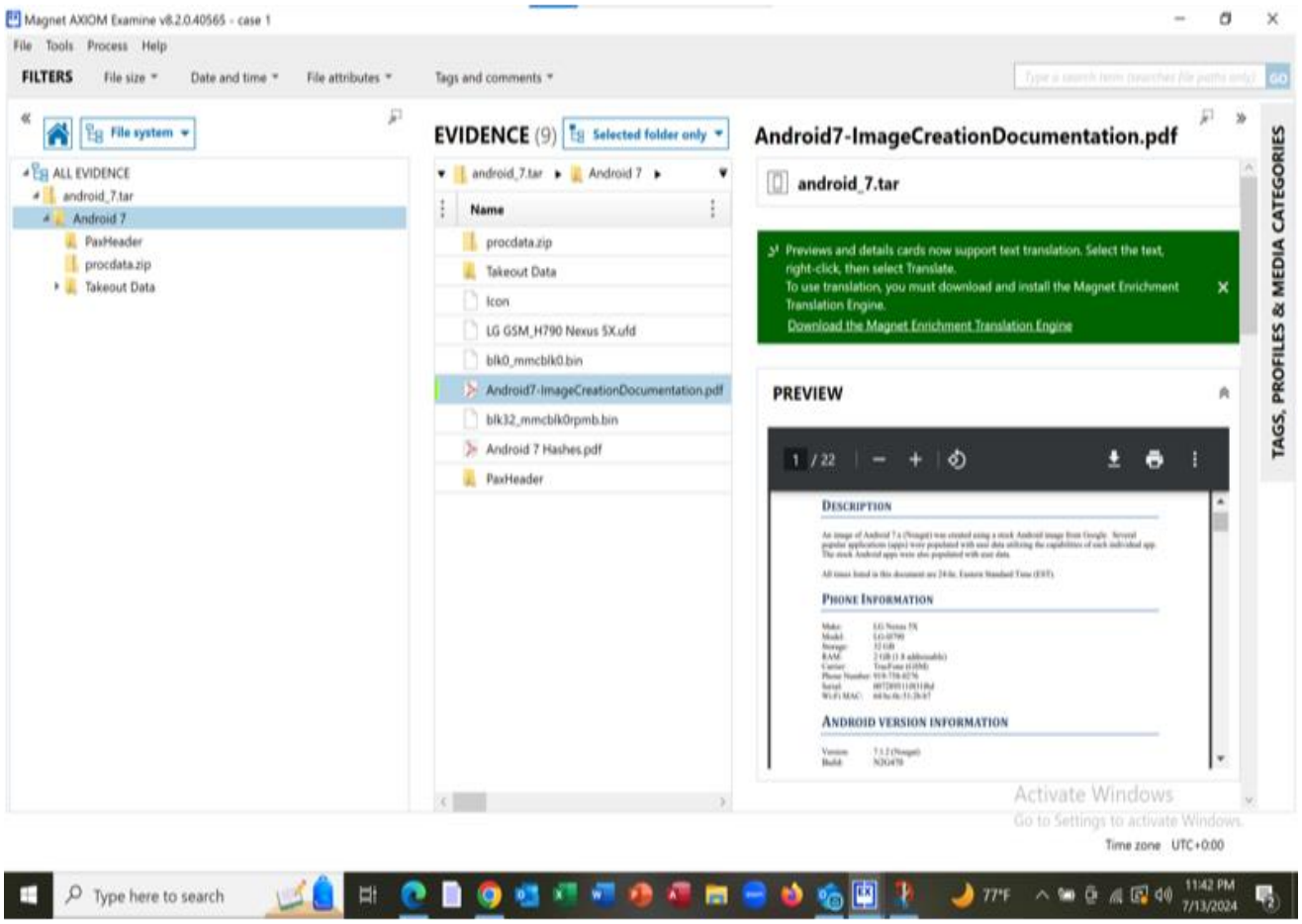
Electronic Media Analyzed

The Media that was analyzed on this device was an Android smartphone. The specifications of this Device will be listed below.

- Make: LG Nexus 5X
- Model: LG-H790
- Storage: 32GB
- RAM: 2 GB (1.8 adressable)
- Carrier: TracFone
- Phone Number: 919-758-0276
- Serial Number: 007289511f431f6d
- Wi-Fi MAC: 64:bc:0c:51:2b:b7
- Android Version: 7.1.2 (Nougat)

This information was obtained through Magnet Axiom's Examine Artifact results by going through the device's File System Information file, accessing the Android 7 subfolder, and accessing the Android7-ImageCreationDocumentation.pdf document. Through this investigation, this

phone's specifications appeared on social media app files within the phone the suspect used and on multiple photos taken on the device's metadata. The screenshot of the device's specifications are shown below.



Report Findings

An Investigative Analysis was performed on the file system from the Android Smartphone device by LG called the Nexus 5X, model LG-H790, serial number 007289511f431f6d. A total of 51,861 artifacts were obtained from this device via Magnet Axiom. Through this investigation, it was determined that this phone was seized and belonged to a Joshua Hickman. This conclusion was made because his name, email, phone number, and phone model were all present in photo metadata, social media profile creations, and internet-based communication applications such as TextNow. This information was discovered in the filesystem of this device. Google account login information was also found in the Android device files. All app activity logs, including usernames, passwords, conversations, and phone numbers, were identified in the Android7-ImageCreationDocumentation.pdf file.

Exhibits

Here will be the Artifact screenshots and documents gathered from this investigation.

Android Nougat Image Image Created by: Joshua Hickman

Android Nougat Image

DESCRIPTION

An image of Android 7.x (Nougat) was created using a stock Android image from Google. Several popular applications (apps) were populated with user data utilizing the capabilities of each individual app. The stock Android apps were also populated with user data.

All times listed in this document are 24-hr, Eastern Standard Time (EST).

PHONE INFORMATION

Make:	LG Nexus 5X
Model:	LG-H790
Storage:	32 GB
RAM:	2 GB (1.8 addressable)
Carrier:	TracFone (GSM)
Phone Number:	919-758-0276
Serial:	007289511f431f6d
Wi-Fi MAC:	64:bc:0c:51:2b:b7

ANDROID VERSION INFORMATION

Version:	7.1.2 (Nougat)
Build:	N2G470
Patch Level:	May 5, 2017
Kernel:	3.10.73-g368902f android-build@kpfj9.cbf.corp.google.com #1 Fri Mar 24 18:12:38 UTC 2017
Baseband:	M8994f-2.6.37.2.21
Passcode:	073116

PROCEDURE

1. The phone was reset to factory defaults, including a wipe of the device. Android was then installed.
2. A new TracFone account was created, and cellular service was applied to the device.
3. A dummy Google account was created and used to manage the device. A secondary Google account was also placed on the phone.
4. Twenty (20) non-stock apps were installed from the Google Play store and populated with user data based on each apps' capabilities.
5. Stock Android apps were populated with user data based on each apps' capabilities.

[!]

GOOGLE ACCOUNT INFORMATION

Email Address:	thisisdfir@gmail.com (default – created on 11/29/2018)
Password:	c%c#W4Qc[tDX;FriUGZP483c2of8&vwC
Email Address:	thisisdfirtwo@gmail.com (created on 12/03/2018)
Password:	patron-maiden-picnic-convene

[!] This Image pertains to the phone information and software information. Also includes the phone's passcode. Google Account information is also included here.

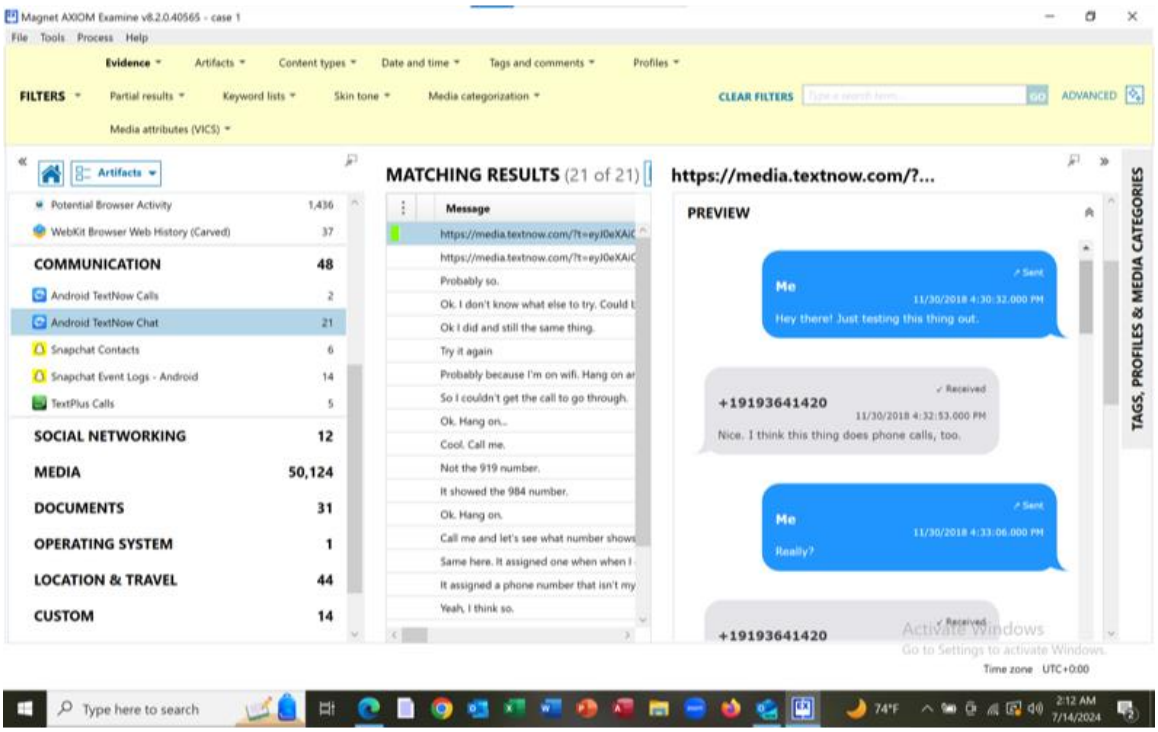
Name: Facebook Messenger
Version Number: 193.0.0
Install Date: 11/30/2018
Install Time: N/A

Username: 919-758-0276
Password: darter-monism-ladylike-disclaim

Date	Time	Action	Message
11/30/2018	14:18	Message Sent	Heeeyyyyyy
	14:19	Message Received	What's up?!
	14:26	Message Sent (x 3)	Sorry, been trying to figure out Tik Tok. The videos are a pain. Have you used it before?
	14:26	Message Received (x 2)	I have not. What is it?
	14:27	Message Sent (x 2)	You post short video clips for people to watch.
			There a lot of lip syncing from what I can tell.
	14:27	Message Received	So, sorta like Snapchat, but not.
	14:27	Message Sent	Yeah, sorta.
	14:28	Message Sent	Emphasis on sorta.
	14:28	Video Received	
	14:29	Message Received	That's the video I posted.
	14:29	Video Sent	
	14:30	Message Sent	That's the video I posted. Not much to look at.
	14:30	Message Received	Mine was lame, too.
	14:31	Picture Received	
	14:31	Message Received	Hahahaha!!!
	14:31	Message Sent (x 2)	Hahahaha!!! That's awesome. We should post it at work.
	14:31	Message Received	I'm on it.
	14:34	Video Call Out (47 seconds)	
	14:36	Audio Call Received (41 seconds)	

[!]

[!] This screenshot is from the Android7-ImageCreationDocumentation.pdf file. Which includes Facebook messenger chat logs and account information.



Name: TextNow
Version Number: 6.4.0
Install Date: 11/29/2018
Install Time: N/A

Username: thisisdfir@gmail.com
Password: $6.Xe6;.q}63AXqaT79N
Texting Address: thisisdfir@textnow.me
TextNow Number: +1 (984) 225-7729

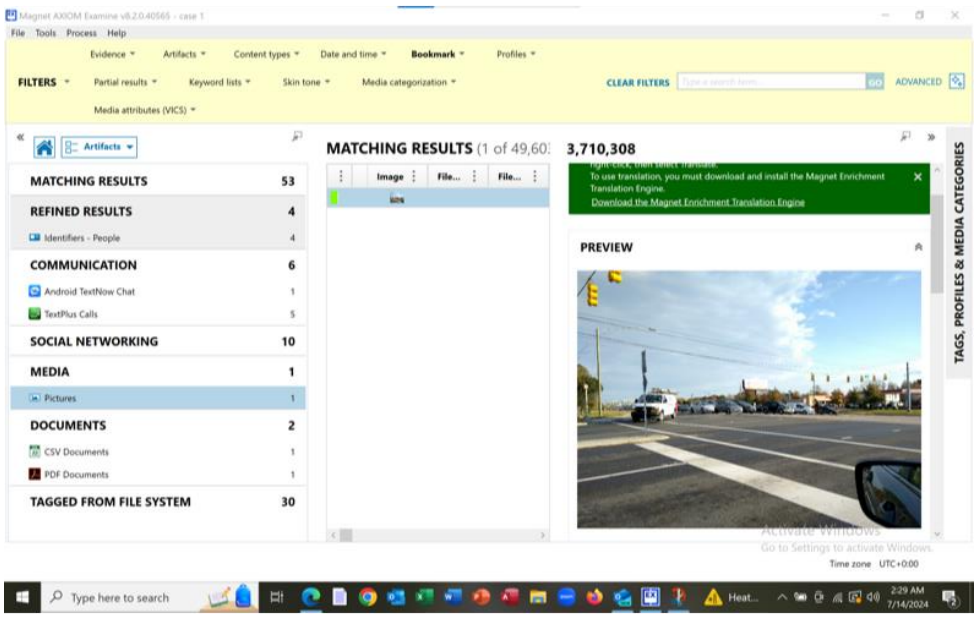
Note:

Date	Time	Action	Messages
11/30/2018	11:30	Message Sent	Hey there! Just testing this thing out.
	11:32	Message Received	Nice. I think this thing does phone calls, too.
	11:33	Message Sent	Really?

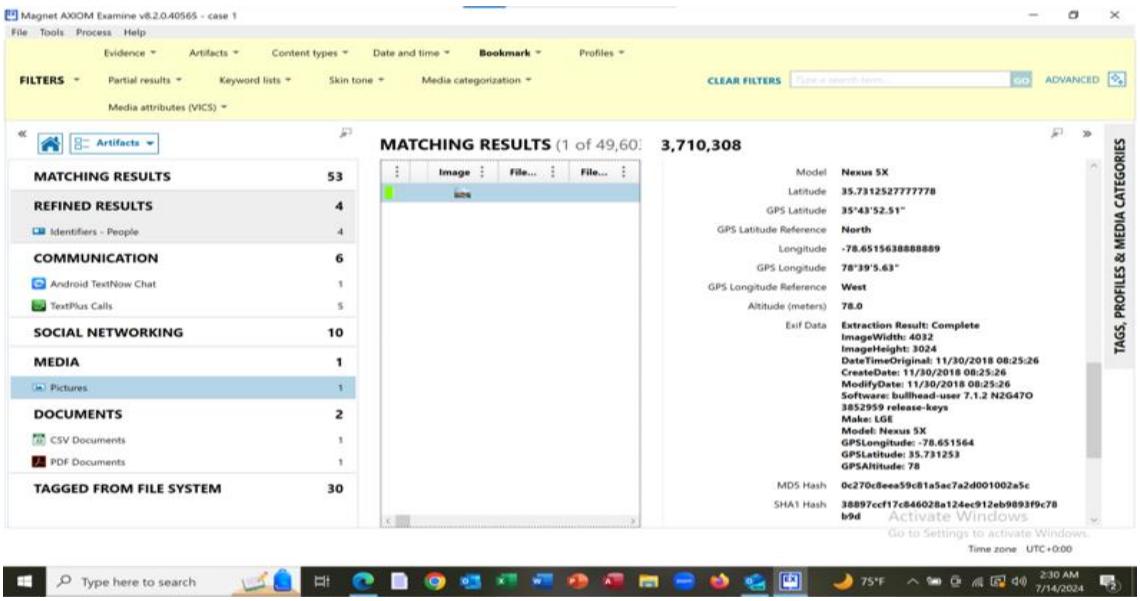
[1] [2]

[1] This is a text now chat from the current phone of how it would appear in the app which was captured by magnet Axiom.

[2] This is a how this very conversation appeared in the phones data log found in the pdf document Android7-ImageCreationDocument.pdf



[3]



[3] Here is a visual of the image taken by this Android phone. On the second picture below the image taken on the phone geolocation information is included about the picture. You are able to see the phone model and longitude and latitude of the location it was taken at.

Glossary

- Personally Identifiable Information (PII) – any form of information that can be used to identify a person either alone or when combined with other information. These forms of information can include emails, names, phone numbers, etc.…
- IoT devices – a network of physical objects that are connected to the internet and can communicate with each other. Examples of IoT devices are smartwatches, smart refrigerators, ring doorbells, health trackers, etc...
- Private Entities – This can be private investigation firms or private investigators.
- Computer Forensics – This is a branch in forensic science that pertains to analyzing or gathering evidence that is related to computers or digital storage media.
- Photo Metadata – data that is embedded in a digital image that is used to describe an image and its production.